

Assignment 4: Fibonacci with memoization

Code

```
def fibonacci(n, memo):
    if n == 0:
        return 0
    if n == 1:
        return 1
    if n in memo:
        return memo[n]
    memo[n] = fibonacci(n - 1, memo) + fibonacci(n - 2, memo)
    return memo[n]

def assignment4():
    n = int(input("Enter n: "))
    memo = {}
    print("fibonacci number:", fibonacci(n, memo))
    print(memo)
```

Output

```
Enter n: 6
fibonacci number: 8
{2: 1, 3: 2, 4: 3, 5: 5, 6: 8}
```